

BUSINESS INTELLIGENCE

Executive Report

Source: test_sales_2000_with_outliers (1).csv
Generated: June 14, 2026 at 07:49

Key Performance Indicators

Metric	Value	Metric	Value
Total Revenue	\$3,107,172.82	MoM Growth	-54.4%
Avg Transaction	\$1,608.27	Next Month Fcst	\$96,900
Top Product	Laptop	Forecast Model	xgboost
Best Region	Central	Data Quality	100/100

1. Executive Summary

The company's total revenue is \$3,107,172.82, with an average transaction value of \$1,608.27. A standout achievement is the top product, Laptop, generating \$941,116.12 in revenue. However, a key concern is the -54.36% month-over-month (MoM) growth, indicating a significant decline in revenue.

2. Key Findings

Total revenue: \$3,107,172.82

Average transaction: \$1,608.27

Top product revenue: \$941,116.12 (Laptop)

Unique SKUs: 6

Best region revenue: \$711,251.24 (Central)

Worst region revenue: \$558,263.97 (East)

Repeat customers: 443

Next month forecast: \$96,900.09

3. Growth Analysis

The company's revenue trajectory is currently declining, with a -54.36% MoM growth and a -2.01% quarter-over-quarter (QoQ) growth. The peak season is September, indicating a potential opportunity to capitalize on seasonal demand. The trend direction is growing steadily, with an expected growth of 62.63% and a confidence level of medium.

The company's compound annual growth rate (CAGR) is -29.76%, which is a concern. However, the forecasting model used, xgboost, has a moderate mean absolute percentage error (MAPE) of 23.22%. This suggests that while there is some uncertainty in the forecast, it is still a useful tool for planning and decision-making.

4. Recommendations

1. Focus on the top 4 products, which generate 80% of revenue, to optimize product offerings and improve profitability.
2. Develop targeted marketing campaigns to improve sales in the worst-performing region, East, which has a revenue of \$558,263.97.
3. Invest in customer retention strategies to increase repeat business from the 443 repeat customers.

4. Review and adjust pricing strategies to reduce the revenue standard deviation of \$1,348.21 and improve average transaction value.
5. Monitor and analyze the performance of the Accessories product, which is the worst-performing product, to identify areas for improvement.

5. Risk Factors

1. Underperforming regions: The East region has a significant performance gap of 21.51% compared to the best-performing Central region, which may impact overall revenue.
2. Concentration risk: The top product, Laptop, generates a significant portion of revenue (\$941,116.12), making the company vulnerable to fluctuations in demand for this product.
3. Forecast uncertainty: The forecasting model has a moderate MAPE of 23.22%, which may lead to inaccurate predictions and impact business decisions, particularly with an expected growth of 62.63% and a trend direction of growing steadily.

Revenue Forecast

Model: xgboost | Accuracy: 76.8% | Confidence: MEDIUM

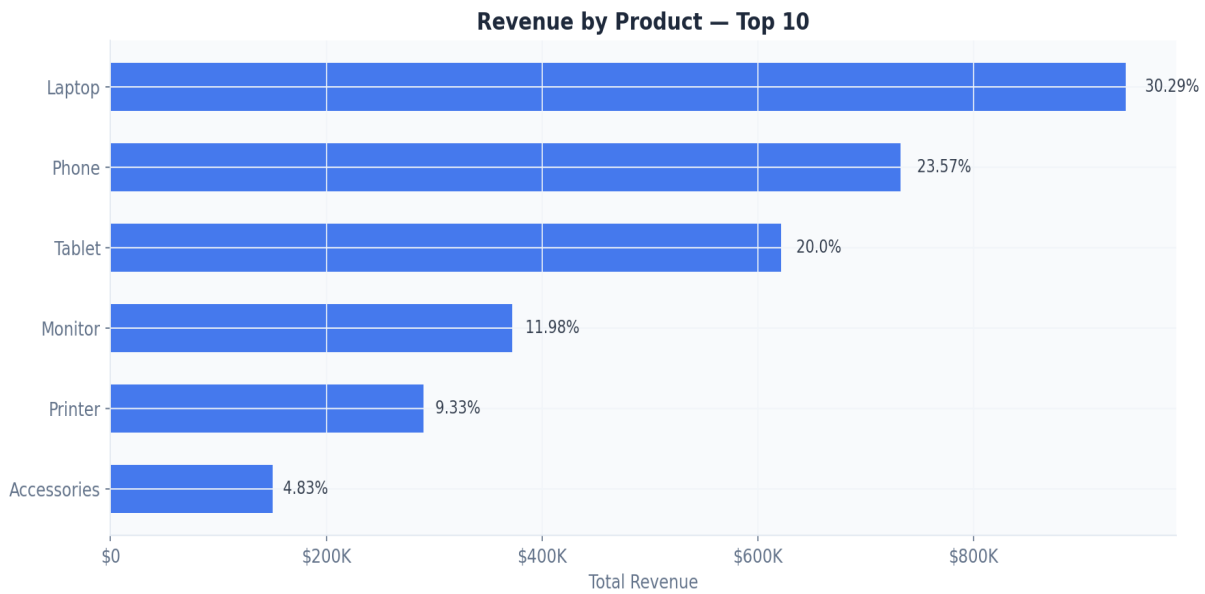
Month	Forecast (\$)	Lower 80%	Upper 80%
2025-07	\$96,900	\$87,210	\$106,590
2025-08	\$112,416	\$101,174	\$123,658
2025-09	\$96,682	\$87,014	\$106,350
2025-10	\$88,464	\$79,618	\$97,311
2025-11	\$105,748	\$95,173	\$116,322
2025-12	\$97,322	\$87,590	\$107,055

Revenue Scenarios

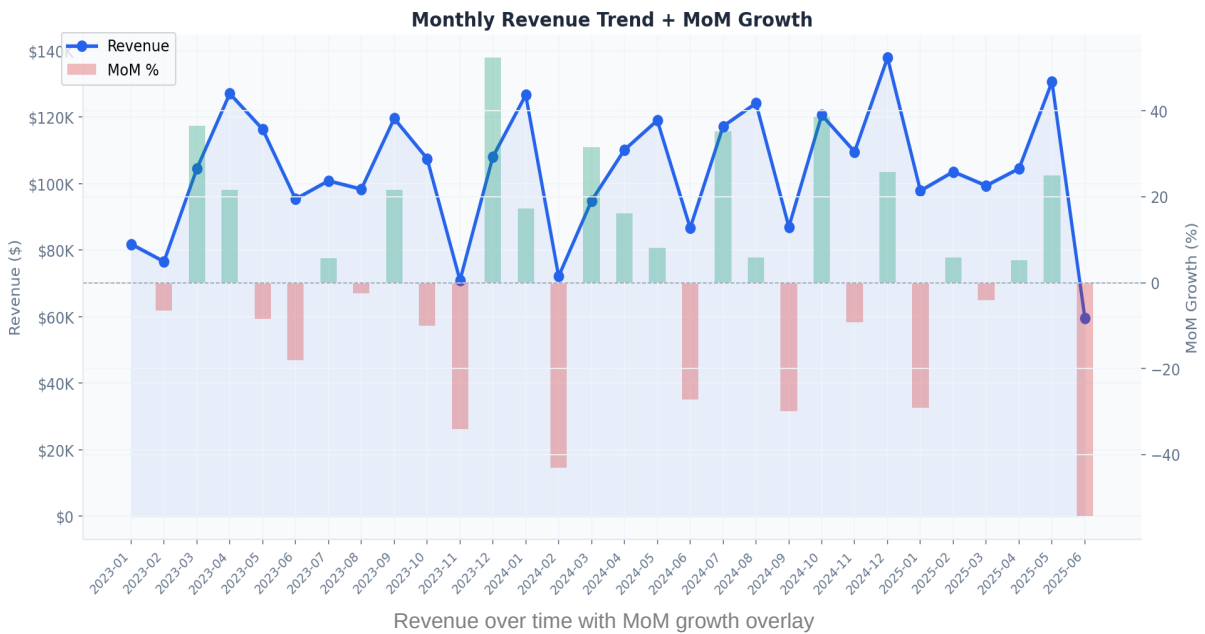
Scenario	Next Month Revenue	Assumption
■ Optimistic	\$116,280	+20% above forecast
■ Realistic	\$96,900	Base forecast
■ Pessimistic	\$77,520	-20% below forecast

Data Visualisations

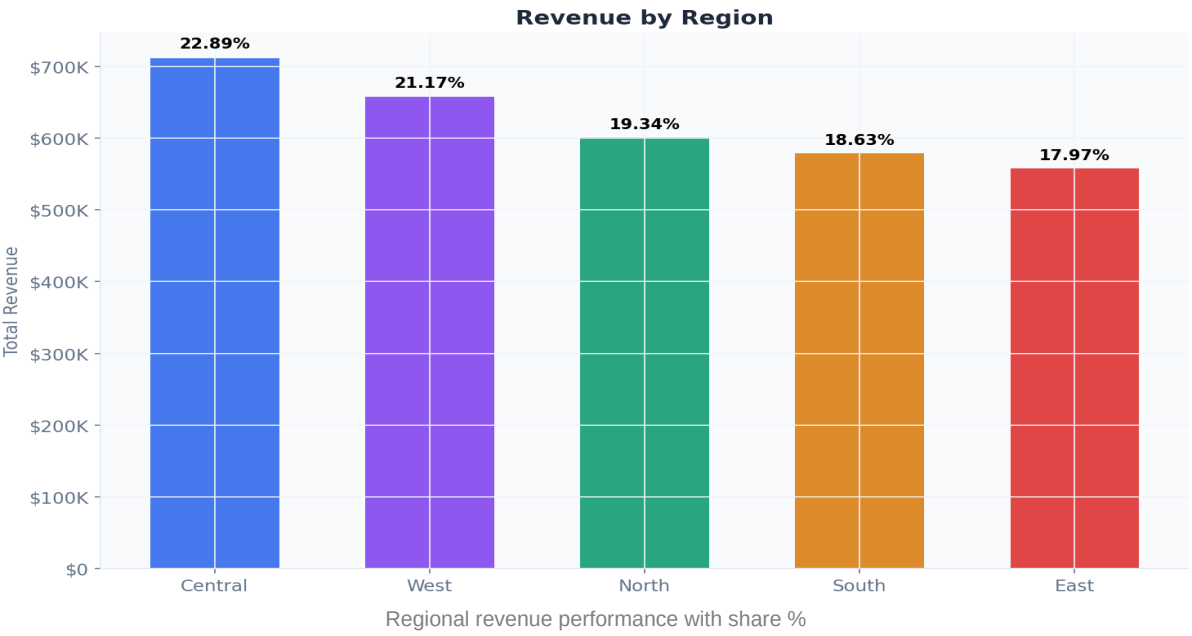
Revenue by Product



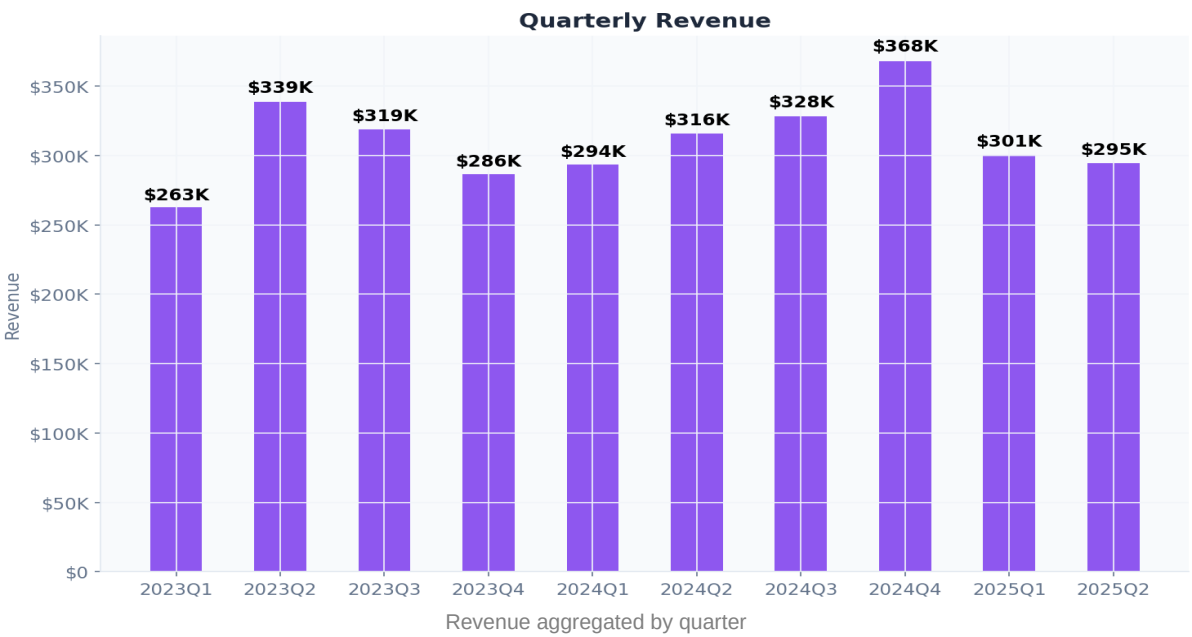
Monthly Revenue Trend



Revenue by Region

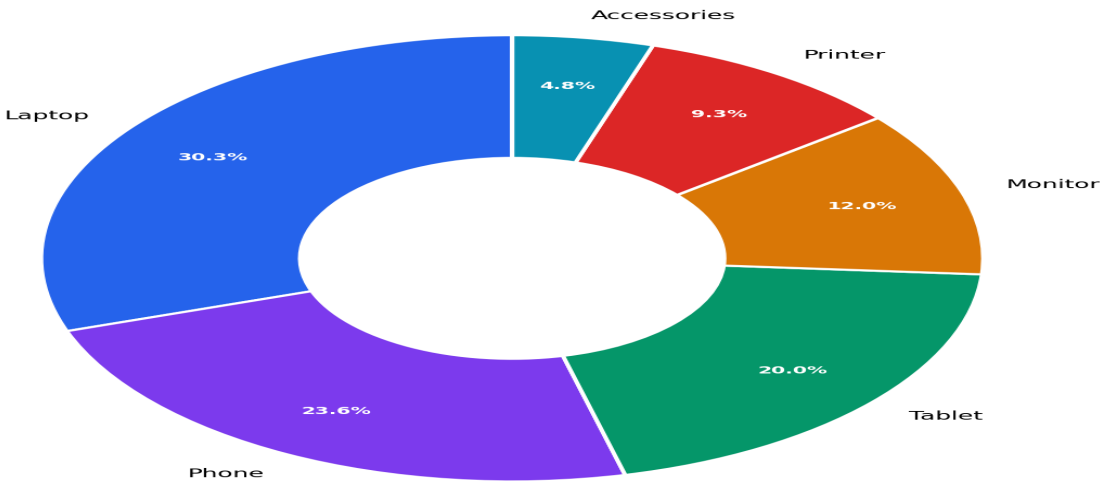


Quarterly Revenue



Revenue Share Donut

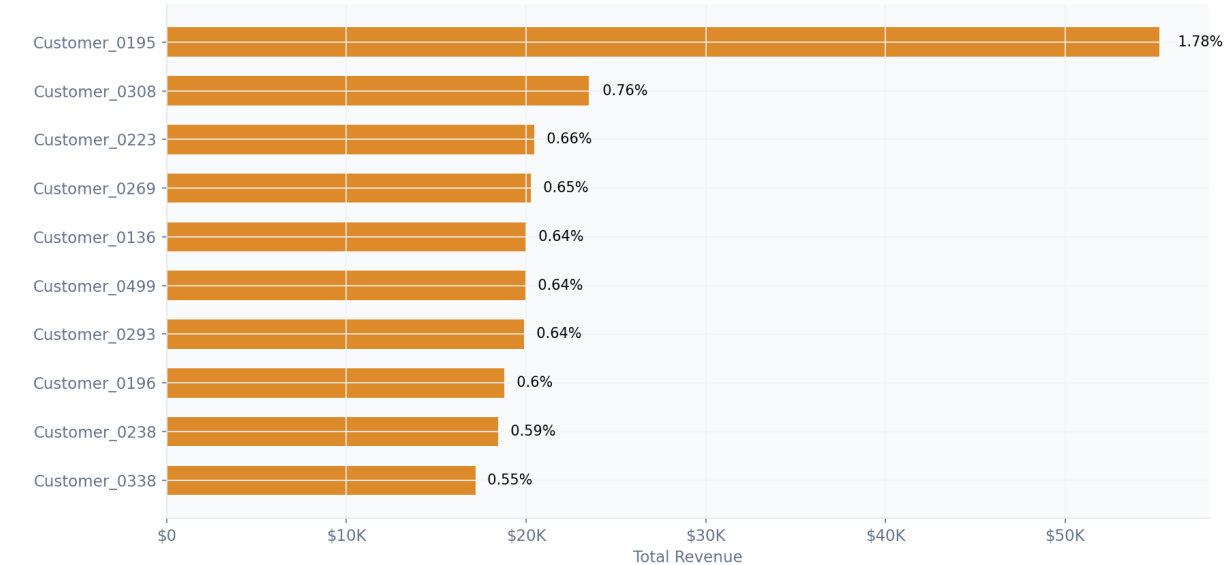
Revenue Share by Product (Top 8)



Revenue share by product (top 8)

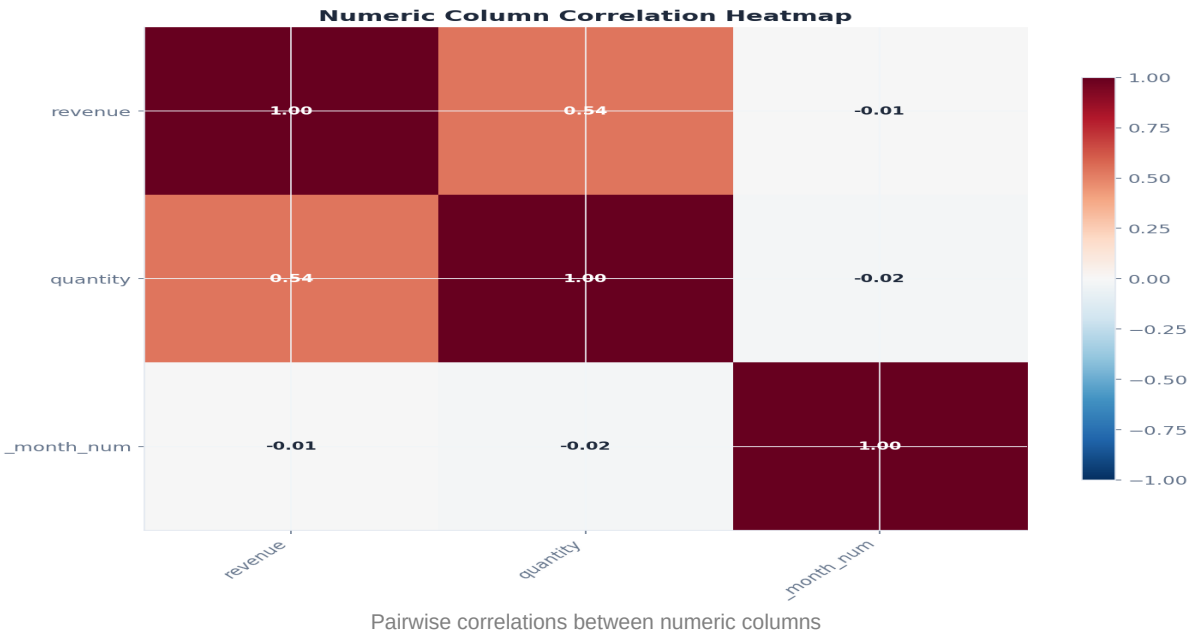
Top 10 Customers

Top 10 Customers by Revenue

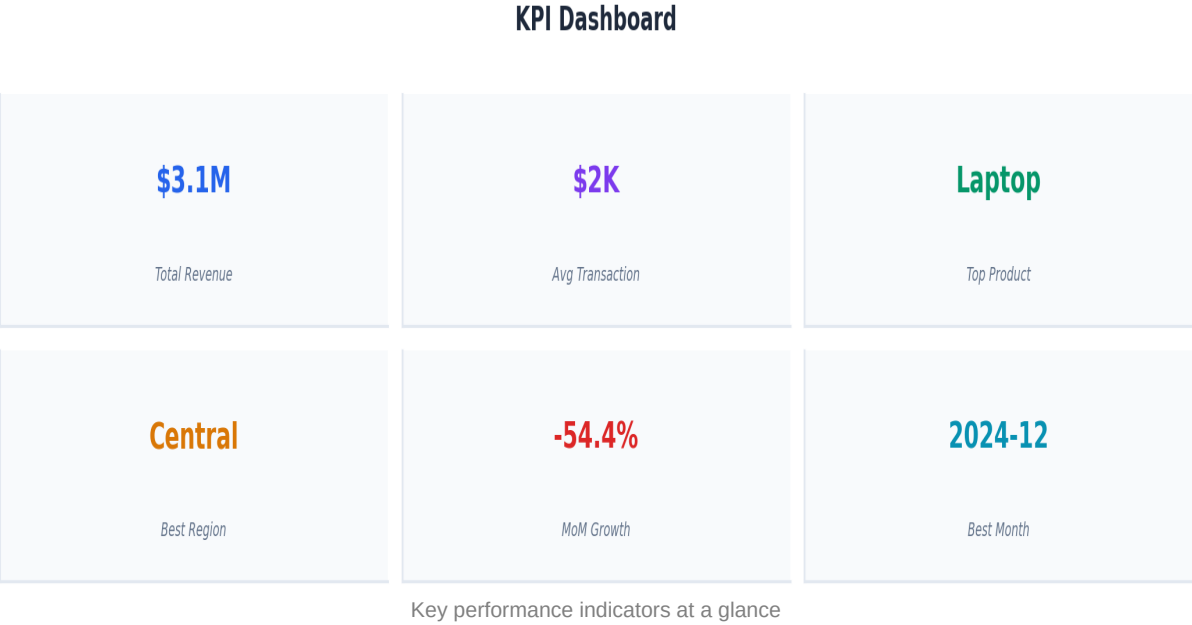


Highest revenue-generating customers

Correlation Heatmap



KPI Dashboard

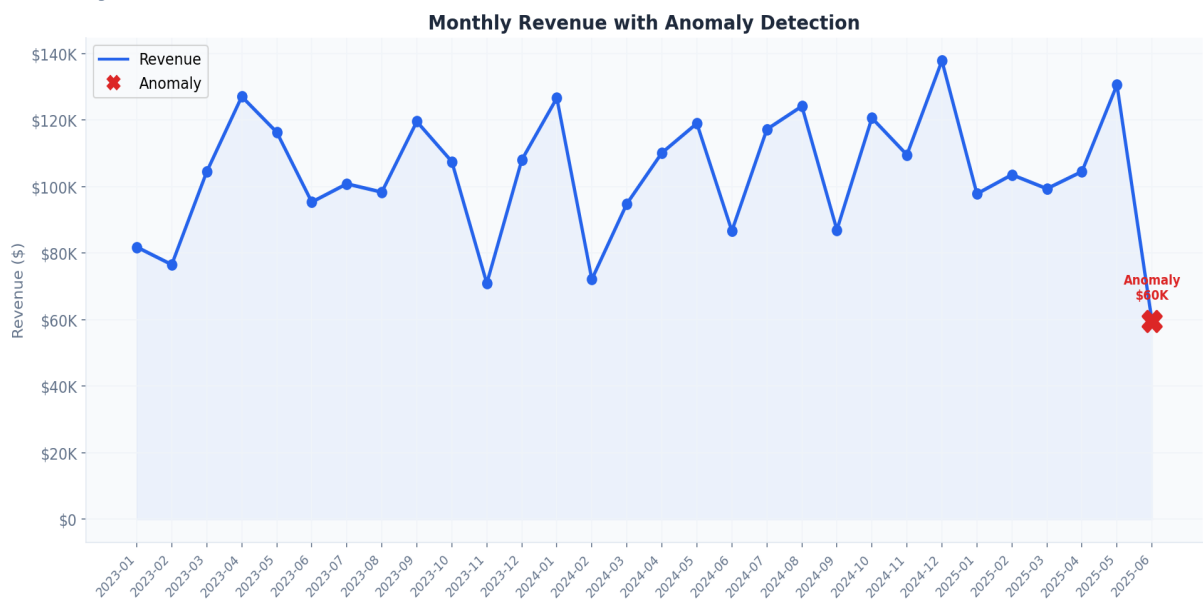


Revenue vs Quantity



Bubble chart — size = revenue, position = qty vs revenue

Anomaly Detection



Months with revenue > 2 std deviations flagged

Appendix: Data Quality Report

Metric	Value
Original Rows	2050
Final Rows	1932
Duplicates Removed	50
Missing Values Fixed	90
Outliers Removed	68
Quality Score	100/100
Date Cols Parsed	date